

AppServer & WebServer

1. Install nginx and run nginx on port number 81.

Step1: install nginx and run nginx

Command Executed:

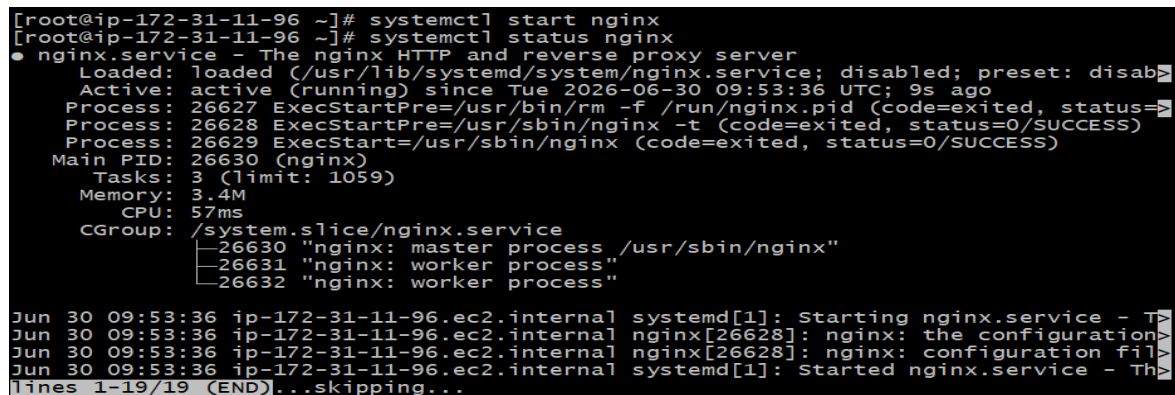
```
sudo yum install nginx
```

```
sudo systemctl enable nginx
```

```
sudo systemctl start nginx
```

```
sudo systemctl status nginx
```

Screenshots:



```
[root@ip-172-31-11-96 ~]# systemctl start nginx
[root@ip-172-31-11-96 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-06-30 09:53:36 UTC; 9s ago
     Process: 26627 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 26628 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 26629 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 26630 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 57ms
    CGroup: /system.slice/nginx.service
            └─26630 "nginx: master process /usr/sbin/nginx"
              └─26631 "nginx: worker process"
                └─26632 "nginx: worker process"

Jun 30 09:53:36 ip-172-31-11-96.ec2.internal systemd[1]: Starting nginx.service - T
Jun 30 09:53:36 ip-172-31-11-96.ec2.internal nginx[26628]: nginx: the configuration
Jun 30 09:53:36 ip-172-31-11-96.ec2.internal nginx[26628]: nginx: configuration fil
Jun 30 09:53:36 ip-172-31-11-96.ec2.internal systemd[1]: Started nginx.service - Th
lines 1-19/19 (END)...skipping...
```

Step 2: change the port number from 80 to 81.

Command Executed:

```
sudo nano /etc/nginx/nginx.conf
```

```
sudo systemctl restart nginx
```

```
netstat -na | grep 81
```

Screenshots:

```
server {
    listen      81;
    listen      [::]:80;
    server_name _;
    root        /usr/share/nginx/html;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;

    error_page 404 /404.html;
    location = /404.html {
    }

    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
    }
}
```

⚠ Not secure 100.55.131.117:81



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

1. Deploy a sample index.html file on nginx.

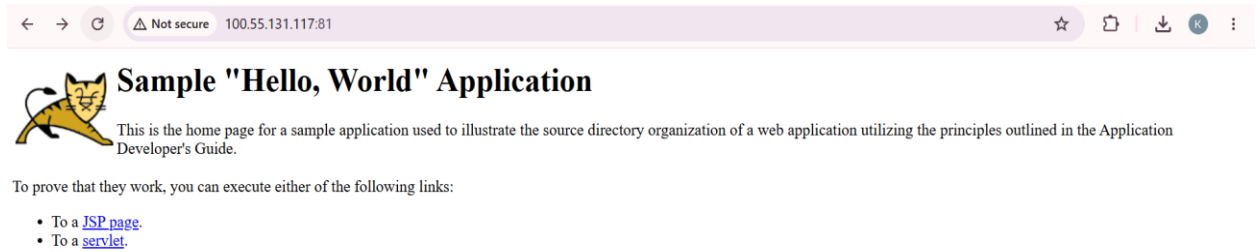
Step1: go to the path of html and replace the index.html with any sample.html.

Commands Executed:

```
cd /usr/share/nginx/html
```

```
sudo nano /usr/share/nginx/html/index.html
```

Screenshots:



2. Install Apache and run Apache on port number 82

Step 1: Install the Apache webserver and run Apache Webserver.

Commands Executed:

```
sudo yum install httpd
```

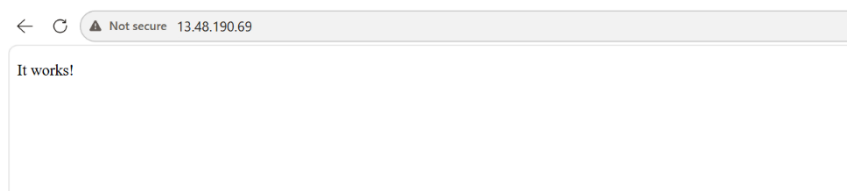
```
sudo systemctl start httpd
```

```
sudo systemctl status httpd
```

Screenshots:

```
[ec2-user@ip-172-31-11-96 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-11-96 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-06-30 11:36:17 UTC; 9s ago
     Docs: man:httpd.service(8)
  Main PID: 29801 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1059)
  Memory: 13.7M
    CPU: 66ms
   CGroup: /system.slice/httpd.service
            └─29801 /usr/sbin/httpd -DFOREGROUND
              └─29802 /usr/sbin/httpd -DFOREGROUND
                └─29803 /usr/sbin/httpd -DFOREGROUND
                  └─29804 /usr/sbin/httpd -DFOREGROUND
                    └─29805 /usr/sbin/httpd -DFOREGROUND

Jun 30 11:36:17 ip-172-31-11-96.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 30 11:36:17 ip-172-31-11-96.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 30 11:36:17 ip-172-31-11-96.ec2.internal httpd[29801]: Server configured, listening on: port 80
```



Step 2: Change the port number from 80 to 82.

Commands Executed:

```
sudo nano /etc/httpd/conf/httpd.conf
```

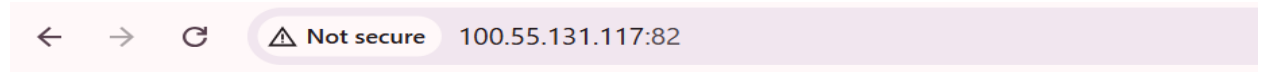
```
sudo systemctl restart httpd
```

Screenshots:



```
#Listen 12.34.56.78:80
Listen 82

#
# Dynamic Shared Object (DSO) Support
#
```



← → ↻ ⚠ Not secure 100.55.131.117:82

It works!

3. Deploy a sample index.html file on Apache.

Step 1: Go to the directory html and Change the index.html According to You.

Commands Executed:

```
cd /var/www/html
```

```
sudo nano /var/www/html/index.html
```

Screenshot:



Welcome to Apache HTTP Server

This is a sample index.html page deployed on Apache.

4. Install Apache tomcat on port number 8082

Step 1: install apache tomcat And run the Apache tomcat.

Commands executed:

```
sudo yum install java-21-amazon-corretto java-21-amazon-corretto -y
```

```
java -version
```

```
sudo nano /etc/profile
```

Add JAVA_HOME path>>

```
export JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto.x86_64
```

```
export PATH=$JAVA_HOME/bin:$PATH
```

<<End.

```
sudo wget https://downloads.apache.org/tomcat/tomcat-10/v10.1.56/bin/apache-tomcat-10.1.56.tar.gz
```

```
sudo tar -xzf apache-tomcat-10.1.56.tar.gz
```

```
sudo mv apache-tomcat-10.1.56 tomcat
```

```
sudo /opt/tomcat/bin/startup.sh
```

```
sudo systemctl status tomcat
```

ScreenShot:

```
[ec2-user@ip-172-31-11-96 ~]$ sudo dnf install java-21-amazon-corretto -y
Last metadata expiration check: 4:44:30 ago on Tue Jun 30 09:52:32 2026.
Dependencies resolved.
```

Package	Arch	Version	Repository	Size
Installing:				
java-21-amazon-corretto	x86_64	1:21.0.11+10-1.amzn2023.1	amazonlinux	218 k
Installing dependencies:				
alsa-lib	x86_64	1.2.7.2-1.amzn2023.0.3	amazonlinux	503 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	467 k
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux	273 k
fonts-filesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
freetype	x86_64	2.13.2-5.amzn2023.0.2	amazonlinux	422 k
giflib	x86_64	5.2.1-9.amzn2023.0.4	amazonlinux	49 k
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k
google-noto-sans-vf-fonts	noarch	20240401-1.amzn2023.0.2	amazonlinux	593 k
graphite2	x86_64	1.3.14-7.amzn2023.0.3	amazonlinux	97 k
harfbuzz	x86_64	7.0.0-2.amzn2023.0.2	amazonlinux	873 k
java-21-amazon-corretto-headless	x86_64	1:21.0.11+10-1.amzn2023.1	amazonlinux	97 M
javapackages-filesystem	noarch	6.0.0-7.amzn2023.0.6	amazonlinux	12 k
langpacks-core-font-en	noarch	3.0-21.amzn2023.0.4	amazonlinux	10 k
libICE	x86_64	1.1.1-3.amzn2023.0.1	amazonlinux	76 k
libSM	x86_64	1.2.4-3.amzn2023.0.1	amazonlinux	45 k
libX11	x86_64	1.8.10-2.amzn2023.0.1	amazonlinux	659 k
libX11-common	noarch	1.8.10-2.amzn2023.0.1	amazonlinux	147 k
libXau	x86_64	1.0.11-6.amzn2023.0.1	amazonlinux	33 k
libXext	x86_64	1.3.6-1.amzn2023.0.1	amazonlinux	42 k
libXi	x86_64	1.8.2-1.amzn2023.0.1	amazonlinux	42 k
libXinerama	x86_64	1.1.5-6.amzn2023.0.1	amazonlinux	16 k
libXrandr	x86_64	1.5.4-3.amzn2023.0.1	amazonlinux	29 k
libXrender	x86_64	0.9.11-6.amzn2023.0.1	amazonlinux	29 k

```
[ec2-user@ip-172-31-11-96 ~]$ java -version
openjdk version "21.0.11" 2026-04-21 LTS
OpenJDK Runtime Environment Corretto-21.0.11.10.1 (build 21.0.11+10-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.11.10.1 (build 21.0.11+10-LTS, mixed mode, sharing)
[ec2-user@ip-172-31-11-96 ~]$
```

```
[ec2-user@ip-172-31-11-96 opt]$ sudo wget https://d1cdn.apache.org/tomcat/tomcat-10/v10.1.56/bin/apache-tomcat-10.1.56.tar.gz
--2026-06-30 14:42:35-- https://d1cdn.apache.org/tomcat/tomcat-10/v10.1.56/bin/apache-tomcat-10.1.56.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14389873 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.56.tar.gz'

apache-tomcat-10.1.56.tar.gz      100%[=====] 13.72M --.-KB/s  in 0.07s

2026-06-30 14:42:36 (199 MB/s) - 'apache-tomcat-10.1.56.tar.gz' saved [14389873/14389873]
```

←
→
×
Not secure
44.212.12.1:8080
☆
🔒
⋮

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Find Help

Apache Tomcat/10.1.56



If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Realms & AAA](#)
- [JDBC DataSources](#)
- [Examples](#)
- [Servlet Specifications](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 10.1 access to the manager application is split between different users.

[44.212.12.1:8080/docs/config/ d more...](#)

Documentation

- [Tomcat 10.1 Documentation](#)
- [Tomcat 10.1 Configuration](#)
- [Tomcat Wiki](#)

Find additional important configuration information in:

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)

Step 2: Change the port number from 8080 to 8082

Command Executed:

```
sudo nano /opt/tomcat/conf/server.xml
```

Find >>

```
<Connector port="8080" protocol="HTTP/1.1"
```

Change it to:

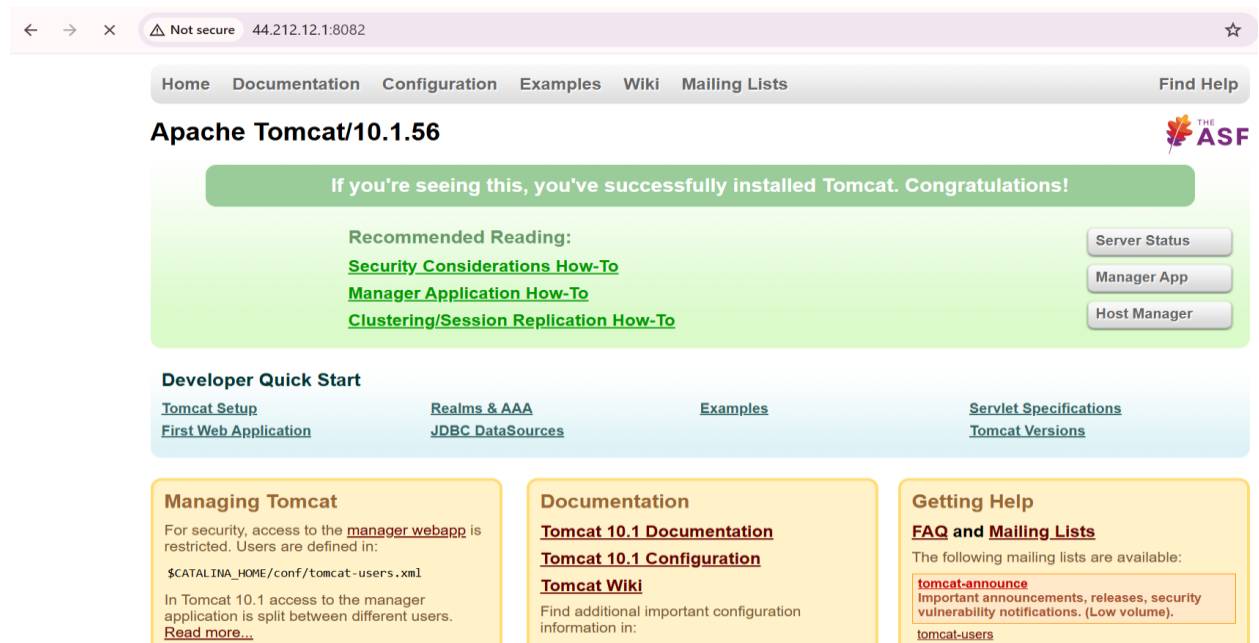
```
<Connector port="8082" protocol="HTTP/1.1"
```

<<End.

```
/opt/tomcat/bin/shutdown.sh
```

```
/opt/tomcat/bin/startup.sh
```

Screenshot:



5. Deploy a sample app on webapps

Step 1: Download Sample Application in webapps folder and unzip the file .

Commands Executed:

```
cd /opt/tomcat/webapps
```

```
wget https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/sample.war
```

```
sudo tar -xzf sample.war.tar.gz
```

Screenshot:



Sample "Hello, World" Application

This is the home page for a sample application used to illustrate the source directory organization of a web application utilizing the principles outlined in the Application Developer's Guide.

To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

6. Create a tomcat.service file for tomcat.

Step 1: Locate the file system directory and create a new directory name service.xml And take service file content from goggle And Replace It.

Commands Executed:

```
sudo vi /etc/systemd/system/tomcat.service
```

```
sudo cat /etc/systemd/system/tomcat.service
```

Screenshot:

```
[ec2-user@ip-172-31-11-96 ~]$ sudo cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking

User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto.x86_64"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

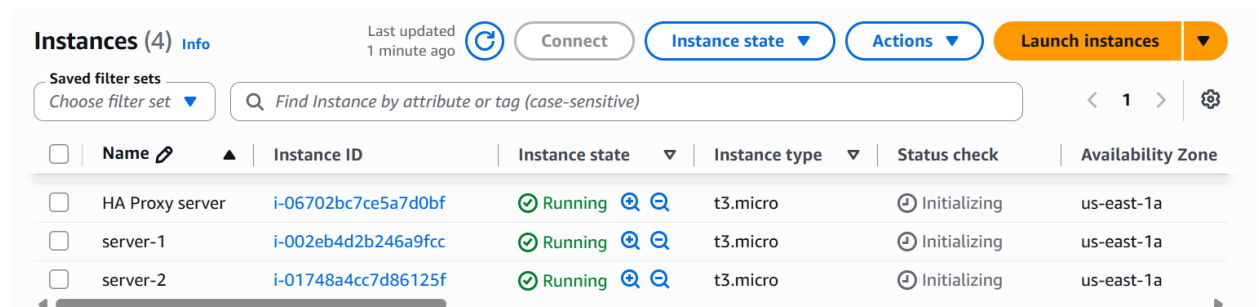
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

7. Configure HA Proxy server

Server-1 Steps

Step1: Launch Three instances with names server-1, server-2 and HA Proxy server.



	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	HA Proxy server	i-06702bc7ce5a7d0bf	Running	t3.micro	Initializing	us-east-1a
☐	server-1	i-002eb4d2b246a9fcc	Running	t3.micro	Initializing	us-east-1a
☐	server-2	i-01748a4cc7d86125f	Running	t3.micro	Initializing	us-east-1a

Step 2: Launch Server1 and go into root user.

Command Executed:

```
ssh -i AppServer.pem ec2-user@public Ip
```

```
sudo -i
```

Screenshot:

```
$ ssh -i AppServer.pem ec2-user@3.231.145.152  
The authenticity of host '3.231.145.152 (3.231.145.152)' can't be established.  
ED25519 key fingerprint is: SHA256:iRaB6OBmmUMlpz05yzzqHl1KWKsd4+1NW3cq9wleQ90  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '3.231.145.152' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html
```

```
#  
#####  
~\ \ #####  
~~ ~\ ####  
~~ ~#/  
~~~~V_/_>  
~~~~_. .  
~/m/_/_/_
```

```
Amazon Linux 2023  
  
https://aws.amazon.com/linux/amazon-linux-2023
```

```
[ec2-user@ip-172-31-8-10 ~]$
```

```

_/_m/'
[ec2-user@ip-172-31-8-10 ~]$ sudo -i
[root@ip-172-31-8-10 ~]#

```

Step 3: Update yum package and install nginx.

Command Executed:

```
sudo yum update
```

```
sudo yum install nginx
```

Screenshot:

```
[root@ip-172-31-8-10 ~]# sudo yum update
Amazon Linux 2023 Kernel Livepatch repository                                453 kB/s | 55 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-8-10 ~]#
```

```

root@ip-172-31-8-10 ~# sudo yum install nginx
Last metadata expiration check: 0:01:48 ago on Wed Jul 1 07:35:18 2026.
Dependencies resolved.

Package                               Architecture      Version           Repository        Size
Installing:
nginx                                 x86_64            1:1.30.2-1.amzn2023.0.1  amazonlinux      34 k
Installing dependencies:
generic-logos-httpd                  noarch            18.0.0-12.amzn2023.0.3  amazonlinux      19 k
gperftools-libs                       x86_64            2.9.1-1.amzn2023.0.3    amazonlinux      308 k
libunwind                             x86_64            1.4.0-5.amzn2023.0.3    amazonlinux      66 k
nginx-core                           x86_64            1:1.30.2-1.amzn2023.0.1  amazonlinux      709 k
nginx-filesystem                     noarch            1:1.30.2-1.amzn2023.0.1  amazonlinux      10 k
nginx-metatypes                       noarch            2:1.49-3.amzn2023.0.3    amazonlinux      21 k

Transaction Summary
Install 7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Is this ok [y/N]: y
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.1 MB/s | 308 kB 00:00
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm      1.4 MB/s | 66 kB 00:00
(3/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 352 kB/s | 19 kB 00:00
(4/7): nginx-1.30.2-1.amzn2023.0.1.x86_64.rpm        1.1 MB/s | 34 kB 00:00
(5/7): nginx-core-1.30.2-1.amzn2023.0.1.x86_64.rpm    19 MB/s | 709 kB 00:00
(6/7): nginx-filesystem-1.30.2-1.amzn2023.0.1.noarch.rpm 305 kB/s | 10 kB 00:00

```

```

Transaction Summary
-----
Install 7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Is this ok [y/N]: y
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm              7.1 MB/s | 308 kB  00:00
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                 1.4 MB/s | 66 kB  00:00
(3/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm     352 kB/s | 19 kB  00:00
(4/7): nginx-1.30.2-1.amzn2023.0.1.x86_64.rpm                   1.1 MB/s | 34 kB  00:00
(5/7): nginx-core-1.30.2-1.amzn2023.0.1.x86_64.rpm              19 MB/s | 709 kB  00:00
(6/7): nginxfilesystem-1.30.2-1.amzn2023.0.1.noarch.rpm         305 kB/s | 10 kB  00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm         703 kB/s | 21 kB  00:00
-----
Total                                                                7.5 MB/s | 1.1 MB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction:
  Preparing :                                                                 1/1
  Running scriptlet: nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch 1/7
  Installing : nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch        1/7
  Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch          2/7
  Installing : libunwind-1.4.0-5.amzn2023.0.3.x86_64                3/7
  Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64         4/7
  Installing : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64           5/7
  Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch    6/7
  Installing : nginx-1:1.30.2-1.amzn2023.0.1.x86_64                7/7
  Running scriptlet: nginx-1:1.30.2-1.amzn2023.0.1.x86_64          7/7
  Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch    1/7
  Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64         2/7
  Verifying : libunwind-1.4.0-5.amzn2023.0.3.x86_64              3/7
  Verifying : nginx-1:1.30.2-1.amzn2023.0.1.x86_64               4/7
  Verifying : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64          5/7
  Verifying : nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch       6/7
  Verifying : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch        7/7

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64      libunwind-1.4.0-5.amzn2023.0.3.x86_64
  nginx-1:1.30.2-1.amzn2023.0.1.x86_64                 nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64        nginxfilesystem-1:1.30.2-1.amzn2023.0.1.noarch
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!

```

Step 5: Locate host file and Add Loadbalancer with public Ip addresses in file
And Ping to that Load balancer.

Command Executed:

sudo nano /etc/hosts

sudo cat /etc/hosts

ping load_balancer -c 4

Screenshot:

```

[root@ip-172-31-8-10 ~]# sudo cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
3.91.14.47  load_balancer

```

```

[root@ip-172-31-8-10 ~]# ping load_balancer -c 4
PING load_balancer (3.91.14.47) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3116ms

```

Step 4: Start the nginx And See the Status of nginx Server.

Commands Executed:

```
sudo systemctl enable nginx
```

```
sudo systemctl start nginx
```

```
sudo systemctl status nginx
```

Screenshot:

```
[root@ip-172-31-8-10 ~]# sudo systemctl start nginx
[root@ip-172-31-8-10 ~]# sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-07-01 07:47:23 UTC; 17s ago
     Process: 26500 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 26507 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 26519 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 26533 (nginx)
       Tasks: 3 (limit: 1059)
      Memory: 3.4M
         CPU: 77ms
    CGroup: /system.slice/nginx.service
            └─26533 "nginx: master process /usr/sbin/nginx"
              └─26535 "nginx: worker process"
                └─26536 "nginx: worker process"

Jul 01 07:47:23 ip-172-31-8-10.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jul 01 07:47:23 ip-172-31-8-10.ec2.internal nginx[26507]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jul 01 07:47:23 ip-172-31-8-10.ec2.internal nginx[26507]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Jul 01 07:47:23 ip-172-31-8-10.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
```

⚠ Not secure 3.231.145.152

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Server-2 Steps:

Step 1: Launch Server2 and go into root user.

Command Executed:

```
ssh -i AppServer.pem ec2-user@public Ip
```

```
sudo -i
```

Screenshot:

Step 3: Locate host file and Add Load_balancer with public Ip addresses in file And Ping to that Load balancer.

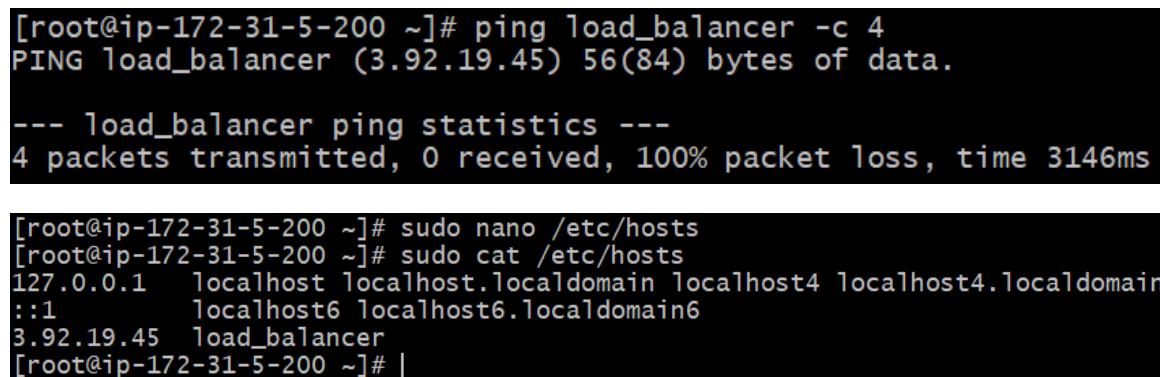
Command Executed:

```
sudo nano /etc/hosts
```

```
sudo cat /etc/hosts
```

```
ping load_balancer -c 4
```

Screenshot:



```
[root@ip-172-31-5-200 ~]# ping load_balancer -c 4
PING load_balancer (3.92.19.45) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3146ms

[root@ip-172-31-5-200 ~]# sudo nano /etc/hosts
[root@ip-172-31-5-200 ~]# sudo cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
3.92.19.45  load_balancer
[root@ip-172-31-5-200 ~]# |
```

Screenshot:

Step 4: Start the nginx And See the Status of nginx Server.

Commands Executed:

```
sudo systemctl enable nginx
```

```
sudo systemctl start nginx
```

```
sudo systemctl status nginx
```

Screenshot:

Step 2: Update yum package and install Haproxy Server.

Command Executed:

```
sudo yum update
```

```
sudo yum install haproxy
```

Screenshot:

```
[root@ip-172-31-8-10 ~]# sudo yum update
Amazon Linux 2023 Kernel Livepatch repository                                453 kB/s | 55 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-8-10 ~]#
```

```
[root@ip-172-31-11-210 ~]# sudo yum install haproxy -y
Last metadata expiration check: 0:02:18 ago on Wed Jul 1 09:12:48 2026.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
haproxy                       x86_64            3.0.5-1.amzn2023.0.1  amazonlinux      2.6 M
=====
Transaction Summary
=====
Install 1 Package
=====
Total download size: 2.6 M
Installed size: 8.2 M
Downloading Packages:
haproxy-3.0.5-1.amzn2023.0.1.x86_64.rpm                                15 MB/s | 2.6 MB    00:00
-----
Total                                                                    11 MB/s | 2.6 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : haproxy-3.0.5-1.amzn2023.0.1.x86_64                1/1
  Running scriptlet: haproxy-3.0.5-1.amzn2023.0.1.x86_64                1/1
  Installing             : haproxy-3.0.5-1.amzn2023.0.1.x86_64                1/1
  Running scriptlet: haproxy-3.0.5-1.amzn2023.0.1.x86_64                1/1
  Verifying               : haproxy-3.0.5-1.amzn2023.0.1.x86_64                1/1
Installed:
haproxy-3.0.5-1.amzn2023.0.1.x86_64
Complete!
[root@ip-172-31-11-210 ~]#
```

Step 3: Locate host file and Add Server1 And Server2 Into Haproxy sever

Command Executed:

```
sudo nano /etc/hosts
```

```
sudo cat /etc/hosts
```

```
ping load_balancer -c 4
```

Screenshot:

```
GNU nano 8.3 /etc/h
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
3.231.145.152 server-1
3.92.19.45  server-2
```

```
[root@ip-172-31-11-210 ~]# sudo nano /etc/hosts
[root@ip-172-31-11-210 ~]# sudo cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
3.231.145.152 server-1
3.92.19.45  server-2
[root@ip-172-31-11-210 ~]# |
```

Step 4: Locate service conf File An Add Front main with port 80 and in BackApp with add publicId:80.

Command Executed:

```
sudo nano /etc/haproxy/haproxy.cfg
```

```
sudo cat /etc/haproxy/haproxy
```

Screenshot:

```

#-----
# main frontend which proxys to the backends
#-----
frontend main
    bind *:80
    acl url_static path_beg          -i /static /images /javascript /stylesheets
    acl url_static path_end          -i .jpg .gif .png .css .js

    use_backend static if url_static
    default_backend app

#-----
# static backend for serving up images, stylesheets and such
#-----
backend static
    balance roundrobin
    server static 127.0.0.1:4331 check

#-----
# round robin balancing between the various backends
#-----
backend app
    balance roundrobin
    server app1 127.0.0.1:5001 check
    server app2 127.0.0.1:5002 check
    server app3 127.0.0.1:5003 check
    server app4 127.0.0.1:5004 check
    server app5 3.231.145.152 check
    server app6 3.92.19.45 check
[root@ip-172-31-11-210 ~]#

```

Step 5: Start the haproxy Server.

Commands Executed:

sudo systemctl enable haproxy

sudo systemctl start haproxy

sudo systemctl status haproxy

Screenshot:

```

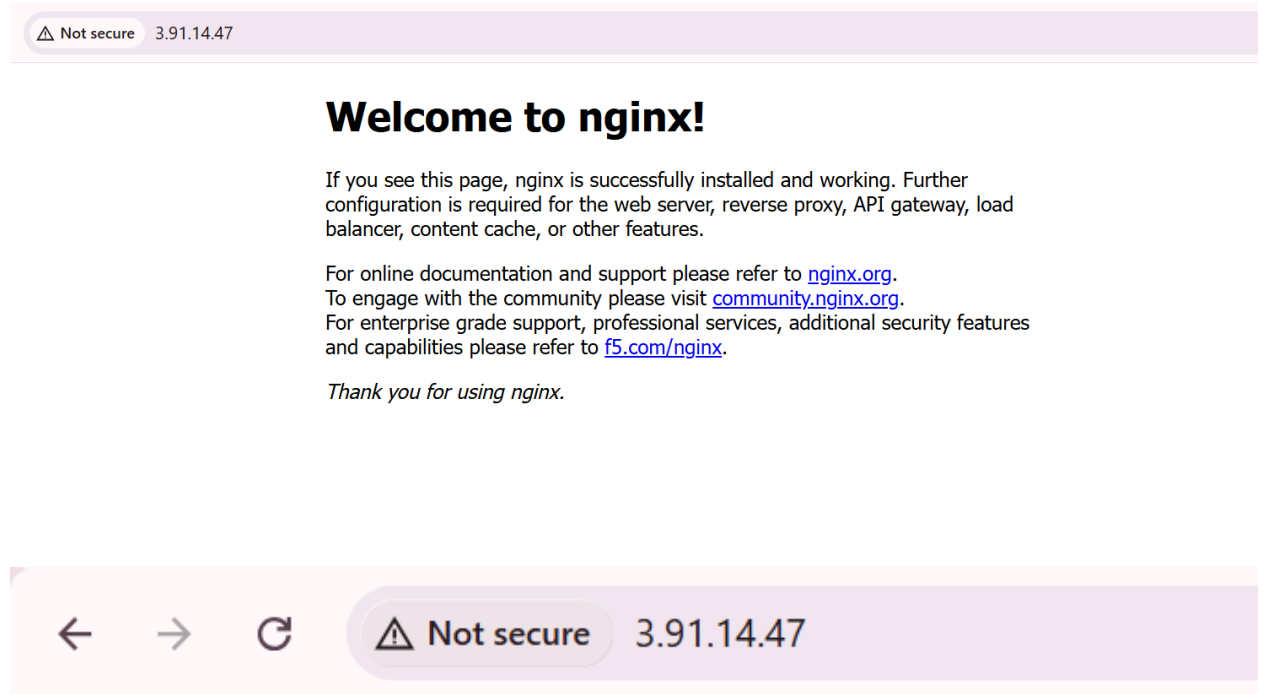
[root@ip-172-31-11-210 ~]# systemctl enable haproxy
[root@ip-172-31-11-210 ~]# systemctl start haproxy
[root@ip-172-31-11-210 ~]# systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-07-01 09:32:54 UTC; 10s ago
     Process: 29173 ExecStartPre=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, status=0/SUCCESS)
    Main PID: 29175 (haproxy)
      Status: "Ready."
        Tasks: 3 (limit: 1059)
       Memory: 7.7M
          CPU: 43ms
    CGroup: /system.slice/haproxy.service
            └─29175 /usr/sbin/haproxy -ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.pid
              29177 /usr/sbin/haproxy -ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.pid

Jul 01 09:32:54 ip-172-31-11-210.ec2.internal haproxy[29175]: [ALERT] (29175) : config : parsing [/etc/haproxy/haproxy.cfg:29] : 'pidfile' already specified. Continuing.
Jul 01 09:32:54 ip-172-31-11-210.ec2.internal haproxy[29175]: [NOTICE] (29175) : New worker (29177) forked
Jul 01 09:32:54 ip-172-31-11-210.ec2.internal haproxy[29175]: [NOTICE] (29175) : Loading success.
Jul 01 09:32:54 ip-172-31-11-210.ec2.internal systemd[1]: Started haproxy.service - HAProxy Load Balancer.
Jul 01 09:32:54 ip-172-31-11-210.ec2.internal haproxy[29177]: [WARNING] (29177) : Server static/static is DOWN, reason: Layer4 connection problem, info: "Connection refused"
Jul 01 09:32:54 ip-172-31-11-210.ec2.internal haproxy[29177]: [ALERT] (29177) : backend 'static' has no server available!
Jul 01 09:32:55 ip-172-31-11-210.ec2.internal haproxy[29177]: [WARNING] (29177) : Server app/app1 is DOWN, reason: Layer4 connection problem, info: "Connection refused"
Jul 01 09:32:55 ip-172-31-11-210.ec2.internal haproxy[29177]: [WARNING] (29177) : Server app/app2 is DOWN, reason: Layer4 connection problem, info: "Connection refused"
Jul 01 09:32:55 ip-172-31-11-210.ec2.internal haproxy[29177]: [WARNING] (29177) : Server app/app3 is DOWN, reason: Layer4 connection problem, info: "Connection refused"
Jul 01 09:32:56 ip-172-31-11-210.ec2.internal haproxy[29177]: [WARNING] (29177) : Server app/app4 is DOWN, reason: Layer4 connection problem, info: "Connection refused"
lines 123/23 (END)

```

Step 5: Now we can Accesses the nginx And Apache webserver With Same public Id in port 80.

Screenshot:



It works!